Chapter 9. Bootstrap Tests

Parametric and nonparametric bootstrap tests, duality with confidence intervals

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The problem

A test needs the law of the statistic *under* H_0, but the bootstrap mimics the *actual* F. Two remedies:

1. **Resample under the null** from an estimate $\hat{F}_0$ satisfying H_0.
2. **Resample a pivot.** If $T_\theta = (\hat{\theta} - \theta)/\widehat{\text{se}}$ is approximately pivotal, approximate its law by $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$ from $\hat{F}$ and compare with T_{θ_0}.

Either way $\hat{p} = \{1 + \#(T^{*b} \text{ at least as extreme as } T_{\text{obs}})\}/(B+1)$.

9.1 Parametric bootstrap tests

In a model $F(\cdot; \theta, \eta)$ with nuisance η, test $H_0 : \theta = \theta_0$ with a statistic $S(\mathbb{X}, \theta_0)$ that is large under H_1.

- *Null resampling:* restricted MLE $\hat{\eta}_0$ under $\theta = \theta_0$; draw from $F(\cdot; \theta_0, \hat{\eta}_0)$, recompute $S(\mathbb{X}^*, \theta_0)$.
- *Pivot resampling:* draw from $F(\cdot; \hat{\theta}, \hat{\eta})$ and recompute $S(\mathbb{X}^*, \hat{\theta})$.

Proposition 9.1

For $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and $S = \sqrt{n}(\bar{X} - \mu_0)/s$, both schemes reproduce the exact null distribution t_{n-1}.

Proof. Null resampling: $N(\mu_0, \hat{\sigma}_0^2)$ samples give $\sqrt{n}(\bar{X}^* - \mu_0)/s^* \sim t_{n-1}$ for any $\hat{\sigma}_0$. Pivot resampling: $N(\hat{\mu}, \hat{\sigma}^2)$ samples give $\sqrt{n}(\bar{X}^* - \hat{\mu})/s^* \sim t_{n-1}$. ■

Exact pivots are rare. For Laplace(μ, b), $2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$ is a pivot only for known b; otherwise use approximate pivots (the t-statistic) or null resampling from Laplace$(\mu_0, \hat{b}_0)$, $\hat{b}_0 = \frac{1}{n}\sum|X_i - \mu_0|$. For two samples with a common scale, fit a common center (pooled median) and a pooled $\hat{b}_0$.

The schemes agree asymptotically; studentized statistics are more accurate than raw differences (Hall and Wilson). Use a non-pivotal statistic (e.g. $\hat{\theta}/\theta_0$ with θ-dependent spread) only with null resampling.

9.2 Nonparametric bootstrap tests

One sample, location. For $H_0 : \theta = \theta_0$ in $X_i = \theta + \epsilon_i$:

- null resampling from the recentered $\tilde{X}_i = X_i - \hat{\theta} + \theta_0$;
- pivot resampling from the raw data with $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$.

For a mean with the t-statistic the two coincide (recentering shifts every bootstrap mean equally). Two-sided: $|T^*| \geq |T_{\text{obs}}|$.

Two samples, $H_0 : F = G$. Draw both samples from the pooled EDF: the with-replacement analogue of the permutation test, similar except in very small samples.

Two samples, equal locations, possibly different shapes. $F \neq G$ is allowed under $H_0 : \mu_1 = \mu_2$, so pooling and permutation are invalid. Use pivot resampling within each sample:

$$T^* = \frac{\bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})}{\sqrt{s_X^{*2}/m + s_Y^{*2}/n}}, \qquad T_{\text{obs}} = \frac{\bar{Y} - \bar{X}}{\sqrt{s_X^2/m + s_Y^2/n}}.$$

Equivalently, recenter each sample to the pooled mean and resample within samples. The *unstandardized* difference of recentered samples without subtracting $\bar{Y} - \bar{X}$ reproduces the sampling law of the observed difference, not a null law.

9.3 Duality with bootstrap intervals

Motivation. Every bootstrap interval yields a test by checking whether θ_0 falls inside.

Proposition 9.2

The two-sided level-α pivot test of $H_0 : \theta = \theta_0$ based on $\hat{\theta} - \theta$ (resp. on $(\hat{\theta} - \theta)/\widehat{\text{se}}$) accepts exactly when θ_0 lies in the basic (resp. bootstrap-t) interval.

Proof. Acceptance means $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta} - \theta_0 \leq Q_{H^*}(1 - \frac{\alpha}{2})$, i.e. $\hat{\theta} - Q_{H^*}(1 - \frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta} - Q_{H^*}(\frac{\alpha}{2})$: (8.1); likewise (8.2). ■

Percentile-type intervals have no simple test counterpart (φ is unknown).

9.4 Selected exercises

Motivation. An un-recentered bootstrap null, an exact Laplace pivot, and two-sample resampling with unequal variances.

Exercise 9.2 (A bootstrap test that never rejects)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\theta, 1)$, treated as coming from an unknown distribution. For $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$, take $S = \bar{X} - \theta_0$, let G^* be the bootstrap distribution of $S^* = \bar{X}^* - \theta_0$ (resampled from the raw data), and use the p-value $p = 1 - G^*(S_{\text{obs}})$. Decide whether: (a) G^* approximates the null distribution of S; (b) the test controls the type I error at $\alpha = 0.05$; (c) the power at $\theta_1 > \theta_0$ is reasonable, also as $\theta_1 \to \infty$.

Solution.

Step 1. (Where G^* is centered.) $S^* = (\bar{X}^* - \bar{X}) + S_{\text{obs}}$, and by Exercise 7.5(a) and the bootstrap CLT $\bar{X}^* - \bar{X} \approx N(0, \hat{\sigma}^2/n)$. So $G^* \approx N(S_{\text{obs}}, \hat{\sigma}^2/n)$ mimics $N(\theta - \theta_0, 1/n)$ around the observed value, not the null $N(0, 1/n)$: (a) fails unless $S_{\text{obs}} \approx 0$.

Step 2. (p does not depend on θ.) $p = P^*(\bar{X}^* > \bar{X})$ is shift invariant, a function of the residuals $x_i - \bar{x}$, whose law under $N(\theta, 1)$ is free of θ. So $P_{\theta_1}(p \leq \alpha) = P_{\theta_0}(p \leq \alpha)$ for every n and θ_1.

Step 3. (Size.) $p = P^*(\sqrt{n}(\bar{X}^* - \bar{X})/\hat{\sigma} > 0) \to \frac{1}{2}$ in probability, so $P(p \leq 0.05) \to 0$.

So (b) holds trivially (the test essentially never rejects) and (c) fails: power equals size for every θ_1. Remedy: recenter at θ_0, or compare the pivot $S^* - S_{\text{obs}} = \bar{X}^* - \bar{X}$ with S_{obs} (Section 9.2). ■

Exercise 9.5 (An exact Laplace pivot)

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu, b)$, with density $\frac{1}{2b}e^{-|x-\mu|/b}$, and b known.
(a) Show that $|X_i - \mu|/b \sim \text{Exp}(1)$ and $V(\mu) = 2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$.
(b) Show that $E|X - c| = d + be^{-d/b}$ with $d = |c - \mu|$, so $E|X - c|$ is uniquely minimized at $c = \mu$.
(c) Conclude that rejecting $H_0 : \mu = \mu_0$ when $V(\mu_0) > q_{1-\alpha}$, the $(1-\alpha)$-quantile of χ^2_{2n}, is an exact level-α test that is consistent against every $\mu \neq \mu_0$.

Solution [Supplement].

(a) With $Y = X - \mu$, $P(|Y|/b > t) = 2\int_{bt}^{\infty} \frac{1}{2b}e^{-y/b}\,dy = e^{-t}$; so $2|X_i - \mu|/b \sim \chi^2_2$ and the sum is χ^2_{2n} (Exercise 1.7(a)).

(b) Take $c = \mu + d$, $d \geq 0$. Use $|Y - d| = (Y - d) + 2(d - Y)^+$, $(d - Y)^+ = \int_{-\infty}^{d} I(Y \leq t)\,dt$, and $P(Y \leq t) = \frac{1}{2}e^{t/b}$ $(t < 0)$, $1 - \frac{1}{2}e^{-t/b}$ $(t \geq 0)$:

$$E(d - Y)^+ = \int_{-\infty}^{0} \tfrac{1}{2}e^{t/b}\,dt + \int_{0}^{d} (1 - \tfrac{1}{2}e^{-t/b})\,dt = d + \tfrac{b}{2}e^{-d/b}.$$

So $E|Y - d| = d + be^{-d/b}$, with derivative $1 - e^{-d/b} > 0$ for $d > 0$.

(c) Under H_0, $P(V(\mu_0) > q_{1-\alpha}) = \alpha$ by (a). Under $\mu \neq \mu_0$, $V(\mu_0)/(2n) \to E|X - \mu_0|/b > 1$ a.s. by the LLN and (b), while $q_{1-\alpha}/(2n) \to 1$; so $P(V(\mu_0) > q_{1-\alpha}) \to 1$. With b unknown $V(\mu_0)$ is not computable, hence Section 9.1. ■

Exercise 9.6 (Two samples with unequal variances)

Let $X_1, \dots, X_m$ have variance σ_X^2 and $Y_1, \dots, Y_n$ variance σ_Y^2, independent, with $H_0 : \mu_X = \mu_Y$.
(a) Recenter both samples at a common value c: $\tilde{X}_i = X_i - \bar{X} + c$, $\tilde{Y}_j = Y_j - \bar{Y} + c$. Show that, drawing the same resampling indices, the studentized statistic computed from $(\tilde{X}^*, \tilde{Y}^*)$ equals the pivot T^* of Section 9.2 computed from (X^*, Y^*).
(b) Show that resampling both samples from the pooled data gives, in large samples, $\text{Var}^*(\bar{Y}^* - \bar{X}^*) \approx \sigma_X^2/n + \sigma_Y^2/m$, whereas $\text{Var}(\bar{Y} - \bar{X}) = \sigma_X^2/m + \sigma_Y^2/n$.

Solution [Supplement].

(a) $\bar{\tilde{X}}^* = \bar{X}^* - \bar{X} + c$, $\bar{\tilde{Y}}^* = \bar{Y}^* - \bar{Y} + c$, so $\bar{\tilde{Y}}^* - \bar{\tilde{X}}^* = \bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})$, and sample variances are shift invariant: the statistics coincide replicate by replicate, for any c.

(b)

Step 1. With $N = m + n$, under H_0 the LLN gives the pooled empirical variance

$$\hat{\sigma}^2_{\text{pool}} \approx \frac{m\sigma_X^2 + n\sigma_Y^2}{N}.$$

Step 2. Both samples are drawn independently from the pooled EDF, so by Exercise 7.5(a)

$$\text{Var}^*(\bar{Y}^* - \bar{X}^*) = \hat{\sigma}^2_{\text{pool}}\Big(\frac{1}{m} + \frac{1}{n}\Big) \approx \frac{m\sigma_X^2 + n\sigma_Y^2}{N} \cdot \frac{N}{mn} = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}.$$

Step 3. The weights of σ_X^2, σ_Y^2 are swapped relative to $\text{Var}(\bar{Y} - \bar{X})$; they agree iff $m = n$ or $\sigma_X^2 = \sigma_Y^2$. Otherwise pooled resampling, like a permutation test, has the wrong null spread (anticonservative when the smaller sample has the larger variance), so resample within samples as in (a). ■